



WORK SHEET - 24

DETERMINANT & MATRICES

Q.1 (D)

$$\Delta_r = \begin{vmatrix} r & r-2 & a \\ 2r & 2r-2 & b \\ 3r & 3r-2 & c \end{vmatrix} = \begin{vmatrix} r & -2 & a \\ 2r & -2 & b \\ 3r & -2 & c \end{vmatrix} = -2r \begin{vmatrix} 1 & 1 & a \\ 2 & 1 & b \\ 3 & 1 & c \end{vmatrix}$$

$$\therefore \sum_{r=1}^n \Delta_r = -2 \cdot \frac{n(n+1)}{2} \begin{vmatrix} 1 & 1 & a \\ 2 & 1 & b \\ 3 & 1 & c \end{vmatrix}$$

$$\therefore \lambda = -n(n+1) \text{ Ans.}$$

Q.2 (A)

$$\frac{d}{dx}(f(f(x))) = f'(f(x))f'(x)$$

$$\left. \frac{d}{dx}(f(f(x))) \right|_{x=0} = f'(f(0))f'(0) = 0 \quad \{ \because f'(0) = 0 \} \text{ Ans.}$$

Q.3 (B)

$$[a, b] \equiv \left[-2, \frac{1}{2} \right]$$

$$a_{ij} \in \left\{ -2, \frac{1}{2}, 4, \frac{1}{4} \right\}$$

$$A = \begin{bmatrix} 4 & 1/2 \\ -2 & 1/4 \end{bmatrix}$$

$$\det. (A) = 1 - (-1) = 2 \text{ which is the least absolute value of } \det.(A).$$

Q.4 (D)

$$M^2 = \mathbf{O}$$

$$\therefore (I + M)^{50} - 50M = I \text{ (Expanding binomially)}$$

Q.5 (A)

$$A^T A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$$|A^T A| = 1 \Rightarrow |A| = \pm 1 \quad \text{But given } |A| = 1$$

$$|A - I| = |A - A^T A| = |A(I - A^T)| = |I - A^T| = |(I - A)^T| = |I - A|$$

$$|A - I| = -|A - I| \Rightarrow |A - I| = 0 \text{ Ans.}$$



Q.6 (C)

Cayley Hamilton Theorem

$$f(x) = x^3 - 5x^2 + 7x - 6$$

$$\left. \begin{array}{l} l = T_r(A) = 5 \\ m = \det(A) = 6 \end{array} \right\} \therefore l + m = 11 \quad \text{Ans.}$$

Q.7 (A)

$$x^3 - 9x^2 + kx - 27 = 0 \quad \begin{array}{l} \alpha \\ \beta \\ \gamma \end{array}$$

$$\left. \frac{\alpha + \beta + \gamma}{3} = \frac{9}{3} = 3 \right\} \Rightarrow \text{A.M.} = \text{G.M.} \Rightarrow \alpha = \beta = \gamma = 3$$

$$(\alpha\beta\gamma)^{1/3} = (27)^{1/3} = 3$$

Hence number of such diagonal matrices is 1. **Ans.**

Q.8 (B)

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\therefore |A| = ad - bc$$

$$x^4 - 13x^3 + px^2 + qx - 64 = 0 \quad \begin{array}{l} a \\ b \\ c \\ d \end{array}$$

Applying A.M. $\geq$ G.M.

$$ad + (-bc) \geq \sqrt{ad(-bc)}$$

$$\frac{ad - bc}{2} \geq \sqrt{64} \Rightarrow ad - bc \geq 16$$

$$\Rightarrow |A| \geq 16 \quad \text{Ans.}$$

Q.9 (A)

Q.10 (A)

(i) $|A - A^T| \neq 0 \Rightarrow A$ must be of matrix of even order.

$\therefore$ order of A is 2.

$$B = \text{adj. } A$$

$$AB = A \text{ adj. } A = |A| I$$

$$\text{adj. } (AB) = \text{adj. } (|A| I) = |A|^{2-1} I = |A| I$$

$$\therefore T_r(\text{adj. } AB) = 2|A| = 6$$

(ii) $\text{adj. } (B^2 A^{-1} B^{-1} A) = \text{adj. } (B^2 (BA)^{-1} A)$

$$BA = (\text{adj. } A)A = |A| I$$

$$(BA)^{-1} = \frac{1}{|A|} I$$



$$\therefore B^2 (BA)^{-1} A = B^2 \cdot \frac{1}{|A|} \cdot A = \frac{B \cdot BA}{|A|} = \frac{B \cdot |A| I}{|A|} = B$$

$$\text{adj.}(B^2(BA)^{-1}A) = \text{adj.} B = \text{adj.}(\text{adj.} A) = |A|^{2-2} A = A$$

Q.11 (D)

Q.12 (D)

Q.13 (C)

$$AB = I = BA$$

$$A(B + A) = 2I$$

$$AB + A^2 = 2I \Rightarrow A^2 = I \Rightarrow A = A^{-1} = B$$

$$|A| = \pm 1$$

$$(i) \quad \det. (4 (\text{adj.} A)^2 + 3 (\text{adj.} B)^2) = \det. (7 (\text{adj.} A)^2)$$

$$= 7^3 \cdot |(\text{adj.} A)^2|$$

$$= 343 |A|^4 = 343$$

$$(ii) \quad P = (A^{-1} + B^{-1}) (\text{adj.} (2A) + \text{adj.} (2B))$$

$$= (2A) (2 \text{adj.} (2A)) = 2(2A) \text{adj.} (2A)$$

$$= 2 \cdot |2A| \cdot I$$

$$= 2 \cdot 2^3 |A| I = \pm 16I$$

$$\therefore T_r(P) = \pm 48$$

$$(iii) \quad T_r(M(x)) = \left| \left(\frac{x^4 + 1}{x^2} \right)^{1/3} \text{adj.} A \right| + 3 + \left| \left(\frac{3}{x^2} \right)^{1/3} \text{adj.} B \right|$$

$$= \frac{x^4 + 1}{x^2} |\text{adj.} A| + 3 + \frac{3}{x^2} |\text{adj.} B|$$

$$= \frac{x^2 + 1}{x^2} + \frac{3}{x^2} + 3 = x^2 + \frac{4}{x^2} + 3$$

$$T_r(M(x)) \geq 4 + 3 = 7$$

Q.14 (C)

Sol. It is obvious.

Q.15 (C)

$$\Delta = \begin{vmatrix} 1 & 3 & 4 \\ 2 & 1 & 3 \\ 4 & 7 & 11 \end{vmatrix}, R_3 \rightarrow R_3 - 2R_1 - R_2 \Rightarrow \Delta = \begin{vmatrix} 1 & 3 & 4 \\ 2 & 1 & 3 \\ 0 & 0 & 0 \end{vmatrix} = 0$$

$$\Delta_x = \begin{vmatrix} 5 & 3 & 4 \\ 3 & 1 & 3 \\ 13 & 7 & 11 \end{vmatrix} = 0 = \Delta_y = \Delta_z \Rightarrow (C) \text{ Ans.}$$

Q.16 (ABCD)

$$a_{ij} = \begin{cases} (j^i + 2j - ij)x, & \text{if } i < j, x \in \mathbb{R} \\ 1, & \text{if } i > j \\ 0, & \text{if } i = j \end{cases}$$

$$a_{12} = (2 + 4 - 2)x = 4x$$

$$a_{23} = (9 + 6 - 6)x = 9x$$

$$a_{13} = (3 + 6 - 3)x = 6x$$

$$a_{21} = a_{31} = a_{32} = 1$$

$$a_{11} = a_{22} = a_{33} = 0$$

$$\therefore A = \begin{bmatrix} 0 & 4x & 6x \\ 1 & 0 & 9x \\ 1 & 1 & 0 \end{bmatrix}$$

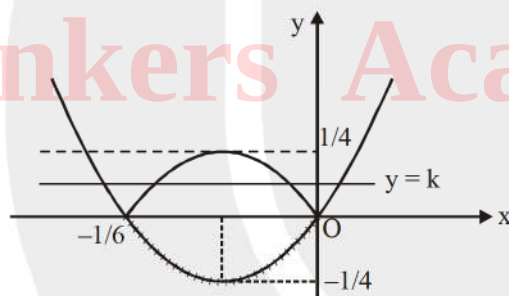
$$|A| = -4x(-9x) + 6x(1) = 36x^2 + 6x$$

$$f(x) = 36x^2 + 6x$$

$$(A) \quad f(x)_{\min} = -\frac{36 - 4 \times 0}{4 \cdot 36} = -\frac{1}{4}$$

$$(B) \quad \int_{-1}^1 f(x) dx = \int_{-1}^1 (36x^2 + 6x) dx = 2 \int_0^1 36x^2 dx = 2 \cdot 36 \cdot \frac{1}{3} = 24$$

$$(C) \quad f(x) = 6x(6x + 1)$$



Clearly, $|f(x)| = k$ has four distinct solutions then $k \in \left(0, \frac{1}{4}\right)$

$$(D) \quad \text{Area} = - \int_{-1/6}^0 (36x^2 + 6x) dx = \left(36 \cdot \frac{x^3}{3} + \frac{6x^2}{2} \right)_{-1/6}^0 = - \left(12x^3 + 3x^2 \right)_{-1/6}^0 = \frac{1}{36}]$$

Q.17 (AD)

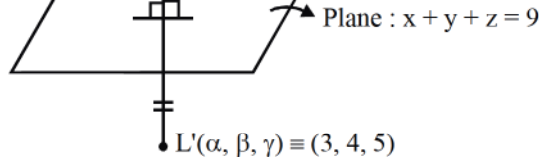
$$AX = C$$

$$\Rightarrow x + 2y + 3z = 14$$

$$4x + y + 2z = 12$$

$$x - y + z = 2$$





$$\therefore L(x=1, y=2, z=3)$$

$$L'(\alpha, \beta, \gamma) \equiv (3, 4, 5)$$

Q.18 [6]

$$[p \ q \ r] \begin{bmatrix} 1 & 9 & 7 \\ 8 & 2 & 7 \\ 7 & 3 & 7 \end{bmatrix} = [0 \ 0 \ 0]$$

$$p + 8q + 7r = 0 \quad \dots (1)$$

$$9p + 2q + 3r = 0 \quad \dots (2)$$

$$7p + 7q + 7r = 0 \quad \dots (3)$$

$$D = \begin{bmatrix} 1 & 8 & 7 \\ 9 & 2 & 3 \\ 7 & 7 & 7 \end{bmatrix} = 0$$

$$D_1 = D_2 = D_3 = 0$$

$\therefore$ infinite solution

$$(3) - (1) \quad 6p - q = 0$$

$$\text{from (3)} \quad p + q + r = 0$$

$$\text{Let } \begin{cases} p = \lambda \\ q = 6\lambda \end{cases} \quad r = -7\lambda$$

$$\therefore M(\lambda, 6\lambda, -7\lambda) \text{ satisfy } \begin{aligned} 2x + y + z &= 1 \\ 2\lambda + 6\lambda - 7\lambda &= 1 \end{aligned}$$

$$\boxed{\lambda = 1}$$

$$\therefore M(1, 6, -7)$$

$$\therefore p = 1, q = 6, r = -7$$

$$\therefore 7p + q + r = 7 + 6 - 7 = 6 \quad \text{Ans.}$$

Q.19 [3]

$$\det. (A) = x_1 \cdot x_2 \cdot x_3$$

$$\det. (B) = y_1 \cdot y_2 \cdot y_3$$

$$\therefore x_1 \cdot x_2 \cdot x_3 = y_1 \cdot y_2 \cdot y_3$$

Both sides

Number of solutions.

$$0 \quad ({}^3C_1 \cdot 2 \cdot 2 + {}^3C_2 \cdot 2 + {}^3C_3)^2 = 19^2 = 361$$

$$1 \quad ({}^3C_2 \cdot 1 + {}^3C_3)^2 = 4^2 = 16$$

$$-1 \quad ({}^3C_2 \cdot 1 + {}^3C_3)^2 = 4^2 = 16$$

$$N = 393$$

$$\therefore \frac{N}{131} = 3. \text{ Ans.}$$

Q.20 [11]

$$(A^3 - 4A^2)(A^2 + 4A + 16I) = \mathbf{0}$$

$$A^2(A - 4I)(A^2 + 4AI + 16I^2) = \mathbf{0}$$

$$A^2(A^3 - 64I^3) = \mathbf{0} \quad [\text{on premultiplying with } (A^{-1})^2 \text{ on both sides, we get}]$$

$$A^3 - 64I = \mathbf{0}$$

$$A^3 = 64I$$

$$|A|^3 = (64)^3$$

$$|A| = 64 = 2^6$$

$$N = |\text{adj. (adj. } 2A)|$$



$$\begin{aligned}
 &= |2A|^{(3-1)^2} = |2A|^4 \\
 &= 2^{12} |A|^4 = 2^{12} \cdot 2^{24} = 2^{36} \\
 \log_{10} N &= 36 \cdot \log_{10} 2 = 36 \times 0.3010 = 10.836 \\
 \therefore \text{number of digits } N &= 11 \quad \text{Ans.}
 \end{aligned}$$

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